

Caoliwen Wang

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Education

University of British Columbia

Sep 2025 - Current

Ph.D. in Computer Science

Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China

Sep 2021 - Jun 2025

B.S. in Computational Mathematics

Thesis Advisor: Prof. Ligang Liu, Prof. Qing Fang

Publications

- [1] Siyuan Chen^{*}, Minghao Guo^{*†}, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen[†]. Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection. *International Conference on Learning Representations (ICLR)*, 2026.
- [2] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang[†]. Neural Shadow Art. *Pacific Graphics (Conference Track)*, 2025.

Preprints

- [1] **Caoliwen Wang**^{*}, Minghao Guo^{*†}, Siyuan Chen^{*}, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen[†]. Unified Simulation of Lagrangian Particle Dynamics via Transformer.
- [2] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang[†]. Shape from Semantics: 3D Shape Generation from Multi-View Semantics.

Teaching

Mathematical Modeling, Teaching Assistant, USTC

Mar 2025 - Jun 2025

Numerical Algebra, Teaching Assistant, USTC

Sep 2024 - Jan 2025

Mathematical Analysis A2, Teaching Assistant, USTC

Mar 2024 - Jul 2024

Introduction to Differential Equations, Teaching Assistant, USTC

Sep 2023 - Jan 2024

Other Research Experience

Graphics&Geometric Computing Laboratory (GCL), USTC

Sep 2023 - June 2025

Advisor: Prof. Juyong Zhang

Geometric Computing and Graphics Group, Cardiff University

June 2024 - June 2025

Advisor: Prof. Bailin Deng

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)